

Agrawal, Gans & Goldfarb (2025)

*The Economics of Bicycles for the Mind*

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Repository 2 - Week of August 24–29, 2026

`github.com/amchavezu/ai-02-agrawal`

**Question.** How do cognitive tools change effort, productivity, and the value of implementation skill and judgment?

**Single mechanism.** A tool substitutes for implementation effort but can complement opportunity and payoff judgment.

Conditional on finding an opportunity, the agent chooses  $e_t \geq 0$ :

$$\max_{e_t \geq 0} M(e_t; \theta) = p(se_t; \theta)\alpha\Delta - c(e_t; \theta).$$

- ▶  $s$ : implementation skill;  $\alpha$ : payoff judgment.
- ▶  $\gamma(t)$ : opportunity judgment;  $\theta$ : tool quality.
- ▶ Interior FOC:  $p'(se_t^*; \theta)s\alpha\Delta = c'(e_t^*; \theta)$ .

**Cognitive-tool conditions.** For  $\theta' > \theta$  and every  $e$ ,

$$p(se; \theta') \geq p(se; \theta), \quad c(e; \theta') \leq c(e; \theta),$$

while  $p'(se; \theta)/c'(e; \theta)$  strictly decreases in  $\theta$  for  $e > 0$ .

With increasing/concave  $p$ , increasing/convex  $c$ , and an interior optimum:

$$e_t^*(1) < e_t^*(0), \quad e_t^*(\theta) = e^*(\theta), \quad V_0(\theta) > V_0(0).$$

*Intuition.* Tool adoption raises achievable net benefit but lowers the marginal return to direct human implementation effort.

- ▶ Mapped the paper's three human inputs: implementation skill, payoff judgment, and opportunity judgment.
- ▶ Reconstructed the per-opportunity optimisation problem and its FOC.
- ▶ Separated Proposition 1 (effort and output), Proposition 2 (adoption drivers), and Proposition 3 (conditional U-shaped variance).
- ▶ Identified the variance claim as the priority audit: the U-shape is conditional and concerns the variance of continuation value  $V(\theta)$ .

**TODO before submission:** add the student's own calculation, numerical check, and final verdict.

**Claim to audit.** “Wage variance is U-shaped in AI quality.”

**Why the loose claim is insufficient.** Proposition 3 requires the paper’s functional form, independence assumptions, positive support, and inequality

$$\frac{\mathbb{E}[\Gamma^2]}{\mathbb{E}[\Gamma]^2} < \mu_s \mathbb{E}\left[\frac{1}{s}\right].$$

It concerns  $\text{Var}(V(\theta))$ , not every possible variance object.

**Verdict:** TODO - write the student’s verdict only after reproducing one step by hand and attaching its photo from hand/.